

Finite difference method

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Problem

Suppose that $S_1(t)$ and $S_2(t)$ are independent stock processes with equal volatility. $S_1(0) = S_2(0) = 100$. The contract at maturity pays the value of the most expensive stock: $V(T) = \max_i\{S_1(T), S_2(T)\}$.

1. What is the formula for the fair value of the contract, $V(0)$?
2. Derive the partial differential equation for this contract

Solution

1. $\max\{S_1(T); S_2(T)\} = S_1(T) + [S_2(T) - S_1(T)]^+$.

The first part's fair prize is $e^{-rT} \mathbf{E}[S_1(T)] = S_1(0) = 100$.

Let the equal volatility be σ . For the second part, use Margrabe's formula: The fair price of $[S_2(T) - S_1(T)]^+$ at $t = 0$ is $S_2(0)N(d_1) - S_1(0)N(d_2)$, where $d_{1,2} = \frac{\ln\left(\frac{S_2(0)}{S_1(0)}\right) \pm \frac{1}{2}\sigma_M^2 T}{\sigma_M \sqrt{T}}$, where $\sigma_M = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2} = \sqrt{2\sigma^2 - 0} = \sqrt{2}\sigma$.

From this: $\ln\left(\frac{S_1(0)}{S_2(0)}\right) = \ln(1) = 0$, so $d_{1,2} = \pm \frac{\sigma\sqrt{T}}{\sqrt{2}}$. So the fair value of the second part is $100N\left(\frac{\sigma\sqrt{T}}{\sqrt{2}}\right) - 100N\left(-\frac{\sigma\sqrt{T}}{\sqrt{2}}\right) = 100N\left(\frac{\sigma\sqrt{T}}{\sqrt{2}}\right) - 100 \cdot \left(1 - N\left(\frac{\sigma\sqrt{T}}{\sqrt{2}}\right)\right) = 200N\left(\frac{\sigma\sqrt{T}}{\sqrt{2}}\right) - 100$.

So the fair price of $V(0)$ is $100 + \left(200N\left(\frac{\sigma\sqrt{T}}{\sqrt{2}}\right) - 100\right) = 200N\left(\frac{\sigma\sqrt{T}}{\sqrt{2}}\right)$.

2. Let $V = V(S_1, S_2, t)$. Applying multivariable Itô-formula to it, we get

$$dV = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S_1} dS_1 + \frac{\partial V}{\partial S_2} dS_2 + \frac{1}{2} \frac{\partial^2 V}{\partial S_1^2} (dS_1)^2 + \frac{1}{2} \frac{\partial^2 V}{\partial S_2^2} (dS_2)^2.$$

The other entires of the formula are 0, because $dt dS_1 = 0$, $(dt)^2 = 0$ and because of the independence of S_1 and S_2 , $dS_1 dS_2 = 0$.

Because S_1 and S_2 are Geometrical Brownian motions,

$$(dS_i)^2 = \sigma^2 S_i^2 dt \quad \forall i \in \{1, 2\},$$

hence

$$\frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S_1} dS_1 + \frac{\partial V}{\partial S_2} dS_2 + \frac{1}{2} \frac{\partial^2 V}{\partial S_1^2} (dS_1)^2 + \frac{1}{2} \frac{\partial^2 V}{\partial S_2^2} (dS_2)^2 = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S_1} dS_1 + \frac{\partial V}{\partial S_2} dS_2 + \frac{1}{2} \sigma^2 \left[\frac{\partial^2 V}{\partial S_1^2} S_1^2 + \frac{\partial^2 V}{\partial S_2^2} S_2^2 \right] dt.$$

Because

$$dS_i = \mu_i S_i dt + \sigma_i S_i dW_i \quad \forall i \in \{1, 2\},$$

we can conclude that

$$\begin{aligned} & \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S_1} dS_1 + \frac{\partial V}{\partial S_2} dS_2 + \frac{1}{2} \sigma^2 \left[\frac{\partial^2 V}{\partial S_1^2} S_1^2 + \frac{\partial^2 V}{\partial S_2^2} S_2^2 \right] dt = \\ &= \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S_1} (\mu_1 S_1 dt + \sigma_1 S_1 dW_1) + \frac{\partial V}{\partial S_2} (\mu_2 S_2 dt + \sigma_2 S_2 dW_2) + \frac{1}{2} \sigma^2 \left[\frac{\partial^2 V}{\partial S_1^2} S_1^2 + \frac{\partial^2 V}{\partial S_2^2} S_2^2 \right] dt = \\ &= \left[\frac{\partial V}{\partial t} + \mu_1 S_1 \frac{\partial V}{\partial S_1} + \mu_2 S_2 \frac{\partial V}{\partial S_2} + \frac{1}{2} \sigma^2 \frac{\partial^2 V}{\partial S_1^2} S_1^2 + \frac{1}{2} \sigma^2 \frac{\partial^2 V}{\partial S_2^2} S_2^2 \right] dt + \sigma_1 S_1 dW_1 + \sigma_2 S_2 dW_2. \end{aligned}$$

Construct a hedged portfolio. $\Pi = V - \frac{\partial V}{\partial S_1} S_1 - \frac{\partial V}{\partial S_2} S_2$. Similarly to the proof shown by Áron Miklós, the non-deterministic entries cancel out in $d\Pi$, and what's left is

$$d\Pi = \left[\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 \frac{\partial^2 V}{\partial S_1^2} S_1^2 + \frac{1}{2} \sigma^2 \frac{\partial^2 V}{\partial S_2^2} S_2^2 \right] dt = r \Pi dt = r \left(V - \frac{\partial V}{\partial S_1} S_1 - \frac{\partial V}{\partial S_2} S_2 \right) dt,$$

which yields that

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S_1^2 \frac{\partial^2 V}{\partial S_1^2} + \frac{1}{2} \sigma^2 S_2^2 \frac{\partial^2 V}{\partial S_2^2} + r S_1 \frac{\partial V}{\partial S_1} + r S_2 \frac{\partial V}{\partial S_2} - r V = 0.$$

with the boundary condition of $V(S_1, S_2, T) = \max\{S_1(T), S_2(T)\}$.